

Three-phase modeling for power flow, optimization, and QSTS.

InterPSS provides a validated static three-phase distribution platform for OpenDSS feeders, unbalanced power-flow studies, DistOPF workflows, and long-horizon quasi-static time-series simulation.

112.3%

of DSS-Python throughput on controlled 8760-step Ckt24 QSTS

InterPSS
1.021 ms/step

DSS-Python
1.147 ms/step

What customers get

OpenDSS-aligned three-phase models

Static parser support for partial-phase feeders, source Thevenin impedance, linecode units, Kron reduction, transformer no-load and magnetizing admittance, OpenDSS load models, PV, storage, regulators, and capacitors.

Power flow and optimization on one network model

Fixed-point unbalanced power flow and DistOPF share static three-phase APIs, avoiding dynamic-model coupling for planning and operational studies.

Validated QSTS for large feeders

Time-series state application, enabled static controls, regulator and capacitor metadata, PV/storage profiles, and measured 8760-step performance.

Measured results

8760

controlled QSTS steps completed on Ckt24

1.021

InterPSS milliseconds per step

0

IEEE8500 voltage/load/branch/generator/capacitor tolerance failures

337

three-phase module tests passing in the latest gate

Static workflow

IMPORT

OpenDSS feeder, device, geometry, profile, and control metadata.

SOLVE

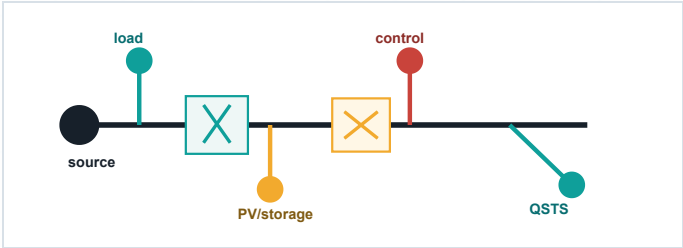
Static three-phase power flow, DistOPF, and QSTS without dynamic models.

VALIDATE

DSS-Python row-level checks for voltage, load, branch, DER, and control outputs.

InterPSS vs. DSS-Python: customer-relevant evidence

EVIDENCE POINT	INTERPSS	DSS-PYTHON REFERENCE	CUSTOMER VALUE
Controlled Ckt24 annual QSTS	1.021 ms/step, 8.948 s total, converged	1.147 ms/step, 10.046 s total, converged	Faster validated annual simulation
IEEE8500 PV-duty voltage parity	Max V delta 0.001047 pu; max angle delta 0.026743 deg	Controlled DSS-Python one-step export	Large-feeder voltage confidence
IEEE8500 branch-flow parity	Max branch P/Q delta 2.874 kW / 2.729 kvar; zero failures	Controlled DSS-Python terminal power export	Operational flow comparison
Static parser and PF gate	337 tests, 0 failures, 0 errors	DSS-Python mini-case and feeder references	Regression-backed delivery



- Model depth:** single-, two-, and three-phase devices with active-phase handling for realistic distribution feeders.
- Planning value:** one static model supports power flow, hosting and DER optimization workflows, and annual time-series studies.
- Validation discipline:** comparisons are grounded in DSS-Python voltages, branch flows, load powers, generator powers, and capacitor states.